

stat_optimal_value_honest_adaptation

Specialization: binary_optimal_value_sparse_neartie_adaptation

TL;DR. The proposal fixes a two-stratum randomized causal union before sampling: an exact-tie Hölder stratum and a rougher margin-constrained near-tie stratum. The ordinary tie-only oracle has root- n interval length. The focal theorem target constructs a sparse randomized-sign full-data near-tie mixture in the second stratum whose product chi-square distance from an exact-tie law remains bounded while its optimal-value separation is $n^{-\xi}$. When $3s_1\alpha < d$, this yields a strictly slower expected-length lower bound at an exact-tie law. At equality the displayed family has root- n order, and above it the family is nonbinding. Ponomarev–Semenova’s one-sided non-uniqueness-robust bands and Liu–Liu–Sasaki–Wan’s Gaussian inference under distinct representative-policy assumptions do not supply this two-sided sparse-mixture lower bound. Cai–Low’s testing reduction is supporting machinery; a matching connected union interval and any stronger obstruction remain open.

1 Project justification

Gap. Existing work provides smoothing, subagging, exceptional-law methods, one-sided bands robust to non-uniqueness, or Gaussian inference under distinct representative-policy assumptions, but not the displayed sparse causal mixture and its two-sided union-honest expected-length cost. Cai–Low supply the general testing reduction, not this causal construction. Armstrong–Kolesar show why an honest upper construction must explicitly pay worst-case bias.

Niche. The kernel isolates an ordered pair for which the oracle exact-tie problem is root- n and, in the explicit regime $3s_1\alpha < d$, a union-honest interval faces a slower, parameterized sparse-near-tie lower bound.

Fill. It supplies dominated full-data laws, verifies their observed images and product testing distance, and states the lower bound as the focal theorem target; a multiscale union-calibrated upper construction is retained only as an open handle.

2 Related work

Luedtke and van der Laan [1] characterize exceptional-law nonregularity and develop optimal-value inference under sufficient conditions. Wu and Wang [2] give smoothed robust and resampling-based optimal-regime inference. Shi, Lu, and Song [3] develop nominal-coverage subagging intervals and compare average lengths. Xu and Guo [4] study adaptive smoothing and a variance bound within a restricted robust asymptotically linear unbiased class. Whitehouse, Chen, Austern, and Syrgkanis [5] develop constant-fit softmax inference for optimal-policy values. Ponomarev and Semenova [6] provide one-sided lower confidence bands robust to non-unique policies through moment-inequality tests, not a connected two-sided interval uniformly asymptotically honest over the displayed ordered union or its sparse-mixture expected-length penalty. Liu, Liu, Sasaki, and Wan [7] give Gaussian optimal-policy-value inference using a strictly convex surrogate under distinct representative-policy assumptions, not the exact-tie ordered union or its two-sided union-honest expected-length lower bound. Cai and Low [8] provide the standard testing-to-interval-length reduction used here as supporting machinery, while the causal sparse mixture is the proposed contribution. Armstrong and Kolesar [9] construct bias-aware honest intervals under predeclared smoothness restrictions and explain why data-driven adaptation needs explicit bias protection; the present finite two-stratum union neither evades that warning nor claims an upper interval before it is proved. Audibert and Tsybakov [10] provide margin and Hölder geometry for the sparse family. Liu, Zhang, Nie, and Soon [11] provide the rilpivirine-versus-efavirenz optimized-regime consumer setting. None of these sources establishes one connected interval uniformly asymptotically honest over this displayed ordered causal union together with its sparse-mixture expected-length penalty.

3 Setup

Target (estimand / structural parameter): $V(P) = E_P[\max\{m_{P,0}(X), m_{P,1}(X)\}]$. Primary functional / estimator / diagnostic: $\delta_n = a_n N_n h_n^d \int_{[0,1]^d} \psi(z) dz / 2 \asymp n^{-2s_1(\alpha+1)/(d+s_1\alpha+4s_1)}$.

- n — `sample_size`; $\mathbb{N}$; index

Definition. Sample size.

- d — `covariate_dimension`; $\mathbb{N}$; `regularity_parameter`

Definition. Covariate dimension.

- B — `outcome_bound`; $(0, \infty)$; constant

Definition. Known outcome bound.

- v_0 — `variance_floor`; $(0, B^2)$; constant

Definition. Known conditional-variance floor.

- $\bar{B}$ — `mean_envelope`; $(0, B)$; `derived_constant`

Definition. $\bar{B} = \sqrt{B^2 - v_0}$.

- $\mathcal{X}$ — `covariate_space`; $[0, 1]^d$; `sample_space`

Definition. Covariate support.

- λ_d — `Lebesgue_measure`; $\mathcal{X}$; `measure`

Definition. Lebesgue measure restricted to $\mathcal{X}$.

- κ_B — `outcome_measure`; $[-B, B]$; `measure`

Definition. $\kappa_B = \lambda_{[-B, B]} + \delta_{-B} + \delta_B$.

- X — `covariate`; $\mathcal{X}$; `causal_primitive`

Definition. Baseline covariate.

- A — `treatment`; $\{0, 1\}$; `causal_primitive`

Definition. Binary treatment.

- $Y(a)$ — `potential_outcome`; $[-B, B]$; `causal_primitive`

Definition. Potential outcome under $a \in \{0, 1\}$.

- Y — `observed_outcome`; $[-B, B]$; `data_component`

Definition. Observed outcome $Y = Y(A)$.

- W — `full_data_unit`; $\mathcal{X} \times \{0, 1\} \times [-B, B]^2$; `data_unit`

Definition. $W = (X, A, Y(0), Y(1))$.

- F — `full_data_law`; $\mathcal{P}(\mathcal{X} \times \{0, 1\} \times [-B, B]^2)$; `causal_model`

Definition. Law of W .

- ν — `full_data_dominating_measure`; $\lambda_d \otimes \#_{\{0, 1\}} \otimes \kappa_B \otimes \kappa_B$; `measure`

Definition. Common dominating measure for the full-data classes.

- T — `observation_map`; $T(x, a, y_0, y_1) = (x, a, y_a)$; `observation_map`

Definition. Measurable observation map.

- O — `observed_data_unit`; $\mathcal{O} = \mathcal{X} \times \{0, 1\} \times [-B, B]$; `data_unit`

Definition. $O = T(W) = (X, A, Y(A))$.

- O_i — `sample_element`; $\mathcal{O}$; `data`

Definition. Observed unit i , with treatment and outcome components A_i and Y_i .

- μ — observed_data_dominating_measure; $\lambda_d \otimes \#_{\{0,1\}} \otimes \kappa_B$; measure
Definition. Common dominating measure for observed-data laws.
- P — observed_data_law; $\mathcal{P}(\mathcal{O})$; model
Definition. $P = T_{\#}F$.
- P_X — covariate_marginal; $\mathcal{P}(\mathcal{X})$; nuisance_law
Definition. Covariate marginal of P .
- g_P — design_density; $\mathcal{X} \rightarrow [0, \infty)$; nuisance
Definition. A fixed Borel version of $dP_X/d\lambda_d$.
- e_P — propensity_score; $\mathcal{X} \rightarrow [0, 1]$; nuisance
Definition. A fixed Borel version of $P(A = 1 \mid X = \cdot)$.
- $m_{\{P,a\}}$ — outcome_regression; $\mathcal{X} \rightarrow [-\bar{B}, \bar{B}]$; nuisance
Definition. A fixed Borel version of $E_P[Y \mid A = a, X = \cdot]$, for $a \in \{0, 1\}$.
- Δ_P — blip_function; $\Delta_P = m_{P,1} - m_{P,0}$; boundary_geometry
Definition. Treatment contrast.
- $V(P)$ — causal_estimand; $\mathbb{R}$; target_estimand
Definition. $V(P) = E_P[\max\{m_{P,0}(X), m_{P,1}(X)\}]$.
- s_0 — tie_smoothness; $(0, \infty)$; regularity_parameter
Definition. Hölder smoothness of the exact-tie stratum.
- L_0 — tie_radius; $(0, \infty)$; regularity_parameter
Definition. Hölder radius of the exact-tie stratum.
- s_1 — near_tie_smoothness; $(0, \infty)$; regularity_parameter
Definition. Hölder smoothness of the near-tie stratum.
- L_1 — near_tie_radius; $(0, \infty)$; regularity_parameter
Definition. Hölder radius of the near-tie stratum.
- α — margin_exponent; $(0, \infty)$; regularity_parameter
Definition. Margin exponent of the near-tie stratum.
- C — margin_constant; $(0, \infty)$; regularity_parameter
Definition. Margin constant of the near-tie stratum.
- β — bump_boundary_power; $(s_1, 1/\alpha)$; construction_parameter
Definition. Fixed boundary power used in the sparse near-tie bump.
- $\mathcal{F}_0$ — exact_tie_full_data_class; $\mathcal{P}(\mathcal{X} \times \{0, 1\} \times [-B, B]^2)$; model_class
Definition. Full-data exact-tie stratum.
- $\mathcal{F}_1$ — near_tie_full_data_class; $\mathcal{P}(\mathcal{X} \times \{0, 1\} \times [-B, B]^2)$; model_class
Definition. Full-data margin stratum.
- $\mathcal{P}_0$ — exact_tie_observed_class; $\mathcal{P}(\mathcal{O})$; model_class
Definition. $\mathcal{P}_0 = \{T_{\#}F : F \in \mathcal{F}_0\}$.

- $\mathcal{P}_1$ — near_tie_observed_class; $\mathcal{P}(\mathcal{O})$; model_class

Definition. $\mathcal{P}_1 = \{T_{\#}F : F \in \mathcal{F}_1\}$.

- $\mathcal{U}$ — union_model; $\mathcal{P}(\mathcal{O})$; adaptation_domain

Definition. $\mathcal{U} = \mathcal{P}_0 \cup \mathcal{P}_1$.

- γ — miscoverage_level; $(0, 1/4)$; inference_constant

Definition. Target miscoverage probability.

- $\mathbf{C}_n$ — generic_interval; $2^{\mathbb{R}}$; decision_rule

Definition. A sample-measurable, almost surely finite connected closed interval $C_n = [L_n, U_n]$ with $L_n \leq U_n$.

- $\mathfrak{C}^{\{\mathrm{asy}\}}(\mathcal{U}, \gamma)$ — honest_interval_sequences; $2^{\{C_n\}_n}$; decision_class

Definition. Sequences satisfying $\liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{U}} P^n\{V(P) \in C_n\} \geq 1 - \gamma$.

- $\mathbf{a}$ — benchmark_sequence; $(0, \infty)^{\mathbb{N}}$; frontier_scale

Definition. Deterministic positive benchmark sequence $a = (a_n)_n$.

- $\mathfrak{R}^{\{\mathrm{asy}\}}_j(\mathbf{a})$ — sequence_minimax_coefficient; $[0, \infty]$; frontier_object

Definition.

$$\mathfrak{R}_j^{\mathrm{asy}}(\mathbf{a}) = \inf_{(C_n)_n \in \mathfrak{C}^{\mathrm{asy}}(\mathcal{U}, \gamma)} \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{P}_j} E_{P^n}[\text{length}(C_n)]/a_n$$

, for $j \in \{0, 1\}$.

- ψ — reference_bump; $\psi(z) = \prod_{r=1}^d [4z_r(1 - z_r)]^{\beta}$; least_favorable_object

Definition. Boundary-vanishing reference bump on $[0, 1]^d$, extended by zero outside that cube.

- D — sparse_mixture_denominator; $(0, \infty)$; derived_constant

Definition. $D = d + s_1\alpha + 4s_1$.

- c_h — cell_width_constant; $(0, \infty)$; calibration_constant

Definition. Fixed cell-width calibration constant.

- c_a — amplitude_constant; $(0, \infty)$; calibration_constant

Definition. Fixed bump-amplitude calibration constant.

- c_N — cell_count_constant; $(0, \infty)$; calibration_constant

Definition. Fixed cell-count calibration constant.

- h_n — cell_width; $(0, 1)$; least_favorable_sequence

Definition. $h_n = c_h n^{-2/D}$, where $c_h > 0$ is a fixed sufficiently small constant.

- a_n — bump_amplitude; $(0, \infty)$; least_favorable_sequence

Definition. $a_n = c_a h_n^{s_1}$, where $c_a > 0$ is fixed.

- N_n — number_of_cells; $\mathbb{N}$; least_favorable_sequence

Definition. $N_n = \lfloor c_N h_n^{s_1\alpha-d} \rfloor$, where $c_N > 0$ is fixed.

- $\mathcal{C}_n$ — cell_collection; $2^{\mathcal{X}}$; least_favorable_object

Definition. The lexicographically first N_n pairwise disjoint cubes of side h_n lying in $[1/4, 3/4]^d$.

- `c_{\ell,n}` — `cell_corner`; $\mathcal{X}$; `least_favorable_object`

Definition. Lower-left corner of the ℓ -th cube in $\mathcal{C}_n$.

- `\theta` — `sign_vector`; $\{-1, 1\}^{N_n}$; `least_favorable_index`

Definition. Rademacher sign vector indexing a sparse near-tie law.

- `\Delta_{\theta,n}` — `sparse_blip`; $\Delta_{\theta,n}(x) = a_n \sum_{\ell=1}^{N_n} \theta_\ell \psi((x - c_{\ell,n})/h_n)$; `least_favorable_object`

Definition. Sparse signed contrast on the cubes $\mathcal{C}_n$.

- `F_{\theta,n}` — `least_favorable_full_data_law`; $\mathcal{F}_1$; `least_favorable_submodel`

Definition. Full-data law constructed from $\Delta_{\theta,n}$ in `def : sparse - near - tie - family`.

- `P_{\theta,n}` — `least_favorable_observed_law`; $\mathcal{P}_1$; `least_favorable_submodel`

Definition. $P_{\theta,n} = T_{\#} F_{\theta,n}$.

- `P_0^{\circ}` — `tie_base_law`; $\mathcal{P}_0$; `least_favorable_submodel`

Definition. Observed image of the zero-mean randomized Rademacher full-data law in `def : base - law`.

- `\overline{Q}_n` — `fuzzy_product_alternative`; $\mathcal{P}(\mathcal{O}^n)$; `testing_object`

Definition. $\overline{Q}_n = 2^{-N_n} \sum_{\theta \in \{-1, 1\}^{N_n}} P_{\theta,n}^n$.

- `\delta_n` — `mixture_value_separation`; $(0, \infty)$; `frontier_scale`

Definition. $\delta_n = a_n N_n h_n^d \int_{[0,1]^d} \psi(z) dz / 2$.

- `\xi` — `adaptation_exponent`; $(0, \infty)$; `derived_rate`

Definition. $\xi = 2s_1(\alpha + 1)/(d + s_1\alpha + 4s_1)$.

- `\varepsilon` — `testing_tolerance`; $(0, 1 - 2\gamma)$; `calibration_constant`

Definition. Fixed testing tolerance.

- `K` — `chi_square_constant`; $(0, \infty)$; `claim_constant`

Definition. Finite constant in the sparse-mixture chi-square bound.

- `c_{\gamma, \varepsilon}` — `length_lower_bound_constant`; $(0, \infty)$; `claim_constant`

Definition. Positive constant depending only on γ and ε .

- `\widehat{V}_{0,n}` — `oracle_tie_estimator`; $\mathbb{R}$; `estimator`

Definition. $\widehat{V}_{0,n} = 2n^{-1} \sum_{i=1}^n 1\{A_i = 0\} Y_i$.

- `C_{0,n}^{\text{or}}` — `oracle_tie_interval`; $2^{\mathbb{R}}$; `decision_rule`

Definition.

$$C_{0,n}^{\text{or}} = [\widehat{V}_{0,n} - B\sqrt{8\log(2/\gamma)/n}, \widehat{V}_{0,n} + B\sqrt{8\log(2/\gamma)/n}]$$

- `C_n^{\dagger}` — `candidate_union_interval`; $2^{\mathbb{R}}$; `decision_rule`

Definition. Connected interval sequence associated with the union-calibrated route.

4 Assumptions

Assumption 1 (ass:iid-sampling). $O_1, \dots, O_n \sim P^n$.

Standard (*iid observed-data sampling*, [1]).

Assumption 2 (ass:consistency). $Y = Y(A)$ under F .

Standard (*consistency*, [1]).

Assumption 3 (ass:exchangeability). $(Y(0), Y(1)) \perp A \mid X \quad [F]$.

Standard (*conditional exchangeability*, [1]).

Assumption 4 (ass:randomization). $e_P(x) = 1/2 \quad \lambda_d\text{-a.e.}$.

Standard (*known randomized propensity*, [1]).

Assumption 5 (ass:bounded-outcomes). $|Y(a)| \leq B \quad F\text{-a.s.}, a \in \{0, 1\}$.

Standard (*bounded potential outcomes*, [1]).

Assumption 6 (ass:variance-floor). $\text{Var}_F(Y(a) \mid X) \geq v_0 \quad F\text{-a.s.}, a \in \{0, 1\}$.

Standard (*nondegenerate conditional variance*, [1]).

Assumption 7 (ass:bounded-density). $1/2 \leq g_P(x) \leq 3/2 \quad \lambda_d\text{-a.e.}$.

Standard (*strong density condition*, [10]).

Assumption 8 (ass:holder-tie). $\max_{a \in \{0, 1\}} \|m_{P,a}\|_{C^{s_0}(\mathcal{X})} \leq L_0$.

Standard (*Hölder regression smoothness*, [10]).

Assumption 9 (ass:holder-near-tie). $\max_{a \in \{0, 1\}} \|m_{P,a}\|_{C^{s_1}(\mathcal{X})} \leq L_1$.

Standard (*Hölder regression smoothness*, [10]).

Assumption 10 (ass:exact-tie). $P_X\{\Delta_P(X) = 0\} = 1$.

Novel. *An exact-treatment-response tie stratum describes settings in which either treatment has the same conditional mean for every baseline profile; it is a low-level benchmark for the exceptional boundary.*

Assumption 11 (ass:margin-near-tie). $P_X\{0 < |\Delta_P(X)| \leq u\} \leq Cu^\alpha \quad \forall u \in (0, 2\bar{B}]$.

Standard (*Tsybakov margin condition with an allowed tie atom*, [10]).

5 Definitions

Definition 1 (def:admissible-pair). Defined object. *admissible ordered pair*

Construction. Write $s = r + \tau$ with $r = \lfloor s \rfloor$ and $\tau \in [0, 1)$; $C^s(\mathcal{X})$ has bounded continuous derivatives through order r , with order- r derivatives τ -Hölder when $\tau > 0$. Fix $0 < s_1 < s_0$, $L_0, L_1 > 0$, and $\alpha, C > 0$ satisfying $s_1\alpha < 1$, and choose $\beta \in (s_1, 1/\alpha)$.

Inputs. $s_0; s_1; L_0; L_1; \backslash \alpha; C; d; \backslash \beta$.

Definition 2 (def:exact-tie-stratum). Defined object. $\mathcal{F}_0$

Construction.

$\mathcal{F}_0 = \{F \ll \nu : F \text{ satisfies } \text{ass} : \text{consistency}, \text{ass} : \text{exchangeability}, \text{ass} : \text{randomization}, \text{ass} : \text{bounded} - \text{outcomes}, \text{ass} : \text{variance} - \text{floor}, \text{ass} : \text{bounded} - \text{density}, \text{ass} : \text{holder} - \text{tie}, \text{ass} : \text{exact} - \text{tie}\}$.

Member properties. *ass:consistency; ass:exchangeability; ass:randomization; ass:bounded-outcomes; ass:variance-floor; ass:bounded-density; ass:holder-tie; ass:exact-tie.*

Definition 3 (def:near-tie-stratum). Defined object. $\mathcal{F}_1$

Construction.

$\mathcal{F}_1 = \{F \ll \nu : F \text{ satisfies } \text{ass} : \text{consistency}, \text{ass} : \text{exchangeability}, \text{ass} : \text{randomization}, \text{ass} : \text{bounded} - \text{outcomes}, \text{ass} : \text{variance} - \text{floor}, \text{ass} : \text{bounded} - \text{density}, \text{ass} : \text{holder} - \text{near} - \text{tie}, \text{ass} : \text{margin} - \text{near} - \text{tie}\}$.

Member properties. *ass:consistency; ass:exchangeability; ass:randomization; ass:bounded-outcomes; ass:variance-floor; ass:bounded-density; ass:holder-near-tie; ass:margin-near-tie.*

Definition 4 (def:observed-strata). Defined object. $\mathcal{P}_0, \mathcal{P}_1$

Construction. $\mathcal{P}_0 = \{T_{\#}F : F \in \mathcal{F}_0\}$ and $\mathcal{P}_1 = \{T_{\#}F : F \in \mathcal{F}_1\}$.

Inputs. T ; $\backslash\textit{mathcal F}_0$; $\backslash\textit{mathcal F}_1$.

Definition 5 (def:union-class). Defined object. $\mathcal{U}$

Construction. $\mathcal{U} = \mathcal{P}_0 \cup \mathcal{P}_1$.

Inputs. $\backslash\textit{mathcal P}_0$; $\backslash\textit{mathcal P}_1$.

Definition 6 (def:base-law). Defined object. *exact-tie base law*

Construction. Let $X \sim \text{Unif}(\mathcal{X})$, let $A \sim \text{Bernoulli}(1/2)$ independently, and conditionally independently draw $Y(0), Y(1) \in \{-B, B\}$ with conditional means zero. Let $\mathcal{P}_0^\circ$ be its observed image.

Inputs. X ; A ; $Y(a)$; B ; T ; $\mathcal{P}_0^\circ \backslash \textit{circ}$.

Definition 7 (def:sparse-near-tie-family). Defined object. *sparse randomized-sign near-tie family*

Construction. For each sufficiently large n , use $\mathcal{C}_n$ and $\Delta_{\theta,n}$. Let $X \sim \text{Unif}(\mathcal{X})$, let $A \sim \text{Bernoulli}(1/2)$ independently, set $E_F[Y(1) \mid X = x] = \Delta_{\theta,n}(x)/2$ and $E_F[Y(0) \mid X = x] = -\Delta_{\theta,n}(x)/2$, and conditionally independently draw $Y(a) \in \{-B, B\}$ with the corresponding Rademacher probabilities. The constants c_h, c_a, c_N are fixed small enough that every displayed mean lies in $[-\bar{B}, \bar{B}]$, every conditional variance is at least v_0 , and the Hölder and margin bounds hold.

Inputs. $\backslash\textit{mathcal C}_n$; $\backslash\Delta_{\theta,n}$; B ; $\bar{B}$; v_0 ; L_1 ; C ; $F_{\theta,n}$.

Definition 8 (def:sequence-frontier). Defined object. *sequence-level honest adaptation criterion*

Construction. For every positive deterministic $a = (a_n)_n$, use $\mathfrak{C}^{\text{asy}}(\mathcal{U}, \gamma)$ and $\mathfrak{R}_j^{\text{asy}}(a)$ exactly as defined in the symbol table; honesty is a property of the entire interval sequence.

Inputs. a ; $\backslash\textit{mathfrak C}^{\{\textit{asy}\}}(\backslash\textit{mathcal U}, \backslash\textit{gamma})$; $\backslash\textit{mathfrak R}^{\{\textit{asy}\}}_j(a)$.

Definition 9 (def:union-upper-route). Defined object. *union-calibrated upper-construction handle*

Construction. Construct a connected interval by fitting multiscale contrast estimates on training folds, using independent holdout folds to form simultaneous confidence bounds for their dyadic block coefficients, and calibrating the final interval to the maximum smoothing and nuisance bias over both $\mathcal{F}_0$ and $\mathcal{F}_1$.

Inputs. $\backslash\textit{mathcal F}_0$; $\backslash\textit{mathcal F}_1$; $\backslash\textit{mathcal U}$; $\mathcal{C}_n$.

6 Main results

Lemma 1 (lem:mean-envelope). Under $\text{ass} : \text{consistency}$, $\text{ass} : \text{exchangeability}$, $\text{ass} : \text{randomization}$, $\text{ass} : \text{bounded} - \text{outcomes}$, and $\text{ass} : \text{variance} - \text{floor}$, consistency, conditional exchangeability, and positivity from known randomization identify $m_{P,a}(x) = E_F[Y(a) \mid X = x]$ for P_X -almost every x and each $a \in \{0, 1\}$; consequently $|m_{P,a}(x)| \leq \bar{B} = \sqrt{B^2 - v_0}$ for P_X -almost every x .

Proof. Fix $a \in \{0, 1\}$. By consistency, on $\{A = a\}$ one has $Y = Y(a)$. Conditional exchangeability and the positivity $P(A = a \mid X) = 1/2$ supplied by known randomization therefore give, for P_X -almost every x ,

$$m_{P,a}(x) = E_P[Y \mid A = a, X = x] = E_F[Y(a) \mid A = a, X = x] = E_F[Y(a) \mid X = x].$$

This proves the asserted identification without defining the causal quantity to be the observed regression.

By bounded outcomes, $E_F[Y(a)^2 \mid X = x] \leq B^2$. The conditional variance identity and $\text{ass} : \text{variance} - \text{floor}$ now imply

$$v_0 \leq \text{Var}_F(Y(a) \mid X = x) = E_F[Y(a)^2 \mid X = x] - E_F[Y(a) \mid X = x]^2 \leq B^2 - m_{P,a}(x)^2.$$

Consequently $m_{P,a}(x)^2 \leq B^2 - v_0 = \bar{B}^2$, and hence $|m_{P,a}(x)| \leq \bar{B}$, for P_X -almost every x . $\square$

Justification. Identification precedes the conditional second-moment calculation that yields the envelope. *Gap.* Both identification and the envelope are standard supporting facts, not contributions. *Consumer.* It supplies the outcome slack used by the sparse family.

Lemma 2 (lem:nonempty-strata). *The base law in $\text{def: base} - \text{law}$ belongs to $\mathcal{F}_0$. For every admissible pair, the zero-contrast law also belongs to $\mathcal{F}_1$; hence both observed-data strata and $\mathcal{U}$ are nonempty.*

Proof. Let F° denote the full-data law specified in $\text{def: base} - \text{law}$. Because X is uniform on $\mathcal{X} = [0, 1]^d$, A has mass $1/2$ at each point of $\{0, 1\}$, and each potential outcome is supported on the two atoms $\{-B, B\}$, the product construction gives $F^\circ \ll \nu$. Indeed, relative to the factors of $\nu = \lambda_d \otimes \#_{\{0,1\}} \otimes \kappa_B \otimes \kappa_B$, the corresponding factor densities are respectively 1, $1/2$, and, for each potential outcome, $1/2$ on each of the atoms $-B$ and B and zero elsewhere.

The observation map gives $Y = Y(A)$, so consistency holds. The conditional mean-zero requirement and the support $\{-B, B\}$ force

$$F^\circ\{Y(a) = B \mid X, A\} = F^\circ\{Y(a) = -B \mid X, A\} = \frac{1}{2}, \quad a \in \{0, 1\}.$$

Thus the conditional law of $(Y(0), Y(1))$ given (X, A) does not depend on A , which proves $(Y(0), Y(1)) \perp A \mid X$. Independence of A and X gives $e_{P_0^\circ}(x) = 1/2$, while uniformity of X gives $g_{P_0^\circ}(x) = 1$, in each case for λ_d -almost every x . Moreover, $|Y(a)| = B$ almost surely and

$$\text{Var}_{F^\circ}(Y(a) \mid X) = B^2 \geq v_0,$$

where the last inequality follows from $v_0 \in (0, B^2)$.

For each $a \in \{0, 1\}$, consistency and the preceding conditional distribution yield

$$m_{P_0^\circ, a}(x) = E_{P_0^\circ}[Y \mid A = a, X = x] = 0$$

for P_X -almost every x . Hence $\|m_{P_0^\circ, a}\|_{C^{s_0}(\mathcal{X})} = 0 \leq L_0$, and $\Delta_{P_0^\circ} = 0$ almost everywhere. All defining conditions in $\text{def: exact} - \text{tie} - \text{stratum}$ therefore hold, so $F^\circ \in \mathcal{F}_0$.

Now fix any admissible pair from $\text{def: admissible} - \text{pair}$ and regard the same F° as the zero-contrast law. The domination, consistency, exchangeability, randomization, boundedness, variance, and density calculations above are unchanged. Since the zero function belongs to every Hölder class with norm zero,

$$\max_{a \in \{0, 1\}} \|m_{P_0^\circ, a}\|_{C^{s_1}(\mathcal{X})} = 0 \leq L_1.$$

For every $u \in (0, 2\bar{B}]$, the event $\{0 < |\Delta_{P_0^\circ}(X)| \leq u\}$ is empty, and consequently

$$P_X\{0 < |\Delta_{P_0^\circ}(X)| \leq u\} = 0 \leq Cu^\alpha.$$

Thus every defining condition in $\text{def: near} - \text{tie} - \text{stratum}$ holds and $F^\circ \in \mathcal{F}_1$.

Finally, by $\text{def: observed} - \text{strata}$, the observed image $P_0^\circ = T_{\#}F^\circ$ belongs to both $\mathcal{P}_0$ and $\mathcal{P}_1$. Hence both observed-data strata are nonempty, and therefore their union $\mathcal{U} = \mathcal{P}_0 \cup \mathcal{P}_1$ is nonempty. $\square$

Justification. A union-honesty problem is meaningful only for compatible dominated full-data classes. *Gap.* This is a model audit rather than a claimed advance over optimal-value procedures. *Consumer.* It validates the ordered pair used in the lower-bound calculation.

Proposition 1 (prop:oracle-tie-rootn). *For every $P \in \mathcal{P}_0$, $E_P[\widehat{V}_{0,n}] = V(P)$, and $P^n\{V(P) \in C_{0,n}^{\text{or}}\} \geq 1 - \gamma$. Consequently $\sup_{P \in \mathcal{P}_0} E_P^n[\text{length}(C_{0,n}^{\text{or}})] = O(n^{-1/2})$.*

Proof. Fix $P \in \mathcal{P}_0$. The exact-tie condition gives $m_{P,1}(X) = m_{P,0}(X)$ almost surely, and thus

$$V(P) = E_P[\max\{m_{P,0}(X), m_{P,1}(X)\}] = E_P[m_{P,0}(X)].$$

Randomization and iterated expectation yield

$$E_P[2\mathbf{1}\{A = 0\}Y \mid X] = 2P(A = 0 \mid X)E_P[Y \mid A = 0, X] = m_{P,0}(X).$$

Averaging over X proves $E_P[\widehat{V}_{0,n}] = V(P)$.

Set $Z_i = 2\mathbf{1}\{A_i = 0\}Y_i$. Under iid sampling, $Z_1, \dots, Z_n$ are independent, have common mean $V(P)$, and lie in $[-2B, 2B]$. Hoeffding's inequality, in its bounded-range form,

$$P^n\left\{\left|n^{-1} \sum_{i=1}^n Z_i - E_P Z_i\right| \geq t\right\} \leq 2 \exp\left\{-\frac{2nt^2}{(4B)^2}\right\} = 2 \exp\left\{-\frac{nt^2}{8B^2}\right\},$$

with $t = B\sqrt{8\log(2/\gamma)/n}$, gives

$$P^n\{V(P) \in C_{0,n}^{\text{or}}\} \geq 1 - \gamma.$$

Finally, the interval length is nonrandom and equals

$$2B\sqrt{8\log(2/\gamma)/n}.$$

Taking the supremum over $P \in \mathcal{P}_0$ proves the claimed $O(n^{-1/2})$ bound. $\square$

Justification. When the contrast is identically zero, the value is an ordinary randomized-arm mean. *Gap.* This is the oracle benchmark, not a novelty claim; Luedtke–van der Laan already motivate regular versus exceptional optimal-value inference. *Consumer.* It defines the root- n precision that union honesty may fail to retain at the exact-tie stratum.

Lemma 3 (lem:sparse-family-membership). *For all sufficiently large n and every $\theta \in \{-1, 1\}^{N_n}$, $F_{\theta,n} \in \mathcal{F}_1$, $P_{\theta,n} \in \mathcal{P}_1$, and $V(P_{\theta,n}) - V(P_0^\circ) = \delta_n$. Moreover $\delta_n \asymp n^{-\xi}$.*

Proof. Write $s = s_1$, let

$$J_1 = \int_{[0,1]^d} \psi(z) dz > 0, \quad J_2 = \int_{[0,1]^d} \psi(z)^2 dz > 0,$$

and extend ψ by zero as in the symbol declaration. Since $\beta > s$, every derivative required by the definition of C^s vanishes continuously at the boundary of the cube, and the order- $\lfloor s \rfloor$ derivatives have the required Hölder modulus. Consequently there is a finite constant M_ψ , depending only on (d, s, β) , such that for every $h \in (0, 1)$, every collection of disjoint cubes $Q_\ell = c_\ell + h[0, 1]^d$, and every signs $\vartheta_\ell \in \{-1, 1\}$,

$$\left\| h^s \sum_{\ell} \vartheta_\ell \psi \left(\frac{\cdot - c_\ell}{h} \right) \right\|_{C^s(\mathcal{X})} \leq M_\psi.$$

Indeed, scaling gives this bound within one cube. Between two cubes, the boundary vanishing of all derivatives through order $\lfloor s \rfloor$ bounds the cross-cube Hölder quotient by the same constant. Disjointness ensures that no point receives two nonzero summands.

We next verify the margin inequality rather than treating it as part of the construction. For $z \in [0, 1]$, put $r(z) = 2\min\{z, 1 - z\}$. Then $4z(1 - z) \geq r(z)$, and $r(Z)$ is uniform on $[0, 1]$ when Z is uniform. Iterated integration gives, for independent uniform $R_1, \dots, R_d$ and $q \in (0, 1]$,

$$\Pr \left\{ \prod_{j=1}^d R_j \leq q \right\} = q \sum_{k=0}^{d-1} \frac{\{\log(1/q)\}^k}{k!}.$$

Because $\alpha\beta < 1$, the right-hand side at $q = t^{1/\beta}$ is at most $M_{\alpha,\beta,d}t^\alpha$, uniformly over $t \in (0, 1]$. The same bound, after increasing the constant, is trivial for $t \geq 1$. Hence

$$\lambda_d\{z \in (0, 1)^d : 0 < \psi(z) \leq t\} \leq M_{\alpha,\beta,d}t^\alpha, \quad t > 0.$$

Since the cubes are disjoint and X is uniform under the construction,

$$\begin{aligned} P_X\{0 < |\Delta_{\theta,n}(X)| \leq u\} &= N_n h_n^d \lambda_d\{0 < \psi \leq u/a_n\} \\ &\leq M_{\alpha,\beta,d} N_n h_n^d (u/a_n)^\alpha \\ &\leq M_{\alpha,\beta,d} c_N c_a^{-\alpha} u^\alpha. \end{aligned}$$

Thus choosing the fixed calibration so that $M_{\alpha,\beta,d}c_Nc_a^{-\alpha} \leq C$ proves *ass : margin – near – tie*.

Now consider the full-data law in *def : sparse – near – tie – family*. Uniform X , randomized A , and endpoint-supported potential outcomes give domination by ν , consistency under the observation map, and conditional exchangeability because A is independent of $(X, Y(0), Y(1))$. They also give $g_P = 1$, $e_P = 1/2$, and bounded outcomes. The observed regressions are

$$m_{P_{\theta,n},1}(x) = \frac{\Delta_{\theta,n}(x)}{2}, \quad m_{P_{\theta,n},0}(x) = -\frac{\Delta_{\theta,n}(x)}{2}.$$

The scaling bound above gives

$$\max_{a \in \{0,1\}} \|m_{P_{\theta,n},a}\|_{C^s(\mathcal{X})} \leq \frac{M_\psi c_a}{2}.$$

Choose $c_a \leq 2L_1/M_\psi$. Also $\|\Delta_{\theta,n}\|_\infty \leq a_n \rightarrow 0$, so eventually $|\Delta_{\theta,n}|/2 \leq \bar{B}$. For the endpoint Rademacher outcome with mean $(2a-1)\Delta_{\theta,n}(x)/2$,

$$\text{Var}_F(Y(a) \mid X = x) = B^2 - \frac{\Delta_{\theta,n}(x)^2}{4} \geq B^2 - \bar{B}^2 = v_0.$$

All conditions defining $\mathcal{F}_1$ now hold, so $F_{\theta,n} \in \mathcal{F}_1$, and its observed image belongs to $\mathcal{P}_1$.

The base law has both regressions equal to zero, hence $V(P_0^\circ) = 0$. Under an alternative,

$$\max\{m_{P_{\theta,n},0}(x), m_{P_{\theta,n},1}(x)\} = \frac{|\Delta_{\theta,n}(x)|}{2}.$$

The signs disappear after taking the absolute value, and a change of variables on every cube gives

$$V(P_{\theta,n}) - V(P_0^\circ) = \frac{1}{2} \int_{\mathcal{X}} |\Delta_{\theta,n}(x)| dx = \frac{a_n N_n h_n^d J_1}{2} = \delta_n.$$

Finally, $s\alpha < 1 \leq d$, so $h_n^{s\alpha-d} \rightarrow \infty$ and

$$N_n \sim c_N h_n^{s\alpha-d}.$$

Therefore

$$\delta_n \asymp h_n^s h_n^{s\alpha-d} h_n^d = h_n^{s(\alpha+1)} \asymp n^{-2s(\alpha+1)/(d+s\alpha+4s)} = n^{-\xi}.$$

□

Justification. The signed construction keeps the causal max functional positive while the signs conceal the alternatives from product testing. *Gap.* Audibert–Tsybakov give classification-margin constructions, but not this full-data randomized causal max-functional family with an exact-tie oracle stratum. *Consumer.* It supplies the explicit cross-stratum alternatives for the adaptation lower bound.

Theorem 1 (*thm:sparse-near-tie-adaptation-lower-bound*). *For every admissible pair satisfying $3s_1\alpha < d$, there are fixed admissible choices of c_h, c_a, c_N and $\varepsilon \in (0, 1-2\gamma)$ such that $\chi^2(\bar{Q}_n, (P_0^\circ)^n) \leq \exp\{Kn^2 N_n a_n^4 h_n^{2d}\} - 1 \leq \varepsilon^2$ eventually. Hence every $(C_n)_n \in \mathfrak{C}^{\text{asy}}(\mathcal{U}, \gamma)$ obeys*

$$\liminf_{n \rightarrow \infty} E_{(P_0^\circ)^n} [\text{length}(C_n)] / \delta_n \geq c_{\gamma,\varepsilon} > 0$$

. Since $3s_1\alpha < d$ gives $\xi < 1/2$, union honesty forces a strictly slower-than-root- n expected length at an exact-tie law even though $\mathcal{P}_0$ alone admits $O(n^{-1/2})$ length.

Proof. Let $R_n = (P_0^\circ)^n$, and write $L_{\theta,n} = dP_{\theta,n}/dP_0^\circ$ for the one-observation likelihood ratio. Under the base law, conditional on (X, A) , the observed outcome is uniform on $\{-B, B\}$. Under $P_{\theta,n}$, its conditional mean is $(2A-1)\Delta_{\theta,n}(X)/2$. Hence

$$L_{\theta,n}(O) = 1 + \frac{(2A-1)Y\Delta_{\theta,n}(X)}{2B^2}.$$

For two sign vectors θ, θ' , independence and symmetry under the base law give

$$E_{P_0^\circ}[L_{\theta,n}L_{\theta',n}] = 1 + \frac{1}{4B^2} \int_{\mathcal{X}} \Delta_{\theta,n}(x)\Delta_{\theta',n}(x) dx = 1 + q_n \sum_{\ell=1}^{N_n} \theta_\ell \theta'_\ell,$$

where

$$q_n = \frac{a_n^2 h_n^d J_2}{4B^2}, \quad J_2 = \int_{[0,1]^d} \psi(z)^2 dz.$$

The cube supports are disjoint, which is exactly what removes all cross-cell integrals.

Squaring the mixture likelihood and using product sampling yields

$$1 + \chi^2(\bar{Q}_n, R_n) = 2^{-2N_n} \sum_{\theta, \theta'} \left(1 + q_n \sum_{\ell=1}^{N_n} \theta_\ell \theta'_\ell \right)^n.$$

If $R_1, \dots, R_{N_n}$ are independent Rademacher variables, the right-hand side equals

$$E \left(1 + q_n \sum_{\ell=1}^{N_n} R_\ell \right)^n.$$

The inner factor is positive because it is an integral of a product of nonnegative likelihoods. Using $1 + x \leq e^x$ and $\cosh t \leq e^{t^2/2}$,

$$\begin{aligned} 1 + \chi^2(\bar{Q}_n, R_n) &\leq E \exp \left(nq_n \sum_{\ell=1}^{N_n} R_\ell \right) \\ &= \{\cosh(nq_n)\}^{N_n} \\ &\leq \exp \left(\frac{N_n n^2 q_n^2}{2} \right) \\ &= \exp \{ K n^2 N_n a_n^4 h_n^{2d} \}, \end{aligned}$$

with $K = J_2^2/(32B^4)$. This proves the first displayed inequality in the theorem.

The calibration constraints used in the membership proof are compatible with the testing constraint. Indeed,

$$n^2 N_n a_n^4 h_n^{2d} \leq c_N c_a^4 n^2 h_n^{d+s_1\alpha+4s_1} = c_N c_a^4 c_h^D.$$

Choose, for example, any fixed $\varepsilon \in (0, 1 - 2\gamma)$, then choose positive c_a, c_N, c_h small enough to satisfy the Hölder, margin, support, and variance calibrations in `def : sparse – near – tie – family` and

$$K c_N c_a^4 c_h^D \leq \log(1 + \varepsilon^2).$$

The collection of N_n cubes exists eventually because $N_n h_n^d \asymp h_n^{s_1 \alpha} \rightarrow 0$, whereas $[1/4, 3/4]^d$ has fixed positive volume. It follows that $\chi^2(\bar{Q}_n, R_n) \leq \varepsilon^2$ eventually.

It remains to transfer testing indistinguishability to interval length. The base law satisfies the exact-tie conditions directly: its regressions vanish, its design density is one, its propensity is one half, and its conditional outcome variances equal B^2 . Thus $P_0^\circ \in \mathcal{P}_0 \subset \mathcal{U}$. By `lem : sparse – family – membership`, every alternative in the mixture belongs to $\mathcal{P}_1 \subset \mathcal{U}$, has value δ_n , and the base value is zero.

Fix any $(C_n)_n \in \mathfrak{C}^{\text{asy}}(\mathcal{U}, \gamma)$. Uniform asymptotic honesty gives a deterministic sequence $r_n \downarrow 0$ such that, eventually,

$$R_n\{0 \in C_n\} \geq 1 - \gamma - r_n, \quad \bar{Q}_n\{\delta_n \in C_n\} \geq 1 - \gamma - r_n.$$

Cauchy–Schwarz gives the total-variation bound

$$\sup_E |\bar{Q}_n(E) - R_n(E)| \leq \frac{1}{2} \left\{ \chi^2(\bar{Q}_n, R_n) \right\}^{1/2} \leq \frac{\varepsilon}{2}.$$

Consequently

$$R_n\{0 \in C_n, \delta_n \in C_n\} \geq 1 - 2\gamma - 2r_n - \frac{\varepsilon}{2}.$$

Because C_n is connected, the event on the left forces $\text{length}(C_n) \geq \delta_n$. Therefore

$$\liminf_{n \rightarrow \infty} \frac{E_{R_n}[\text{length}(C_n)]}{\delta_n} \geq 1 - 2\gamma - \frac{\varepsilon}{2} =: c_{\gamma, \varepsilon} > 0.$$

Finally,

$$\xi - \frac{1}{2} = \frac{3s_1\alpha - d}{2(d + s_1\alpha + 4s_1)}.$$

Thus $3s_1\alpha < d$ implies $\xi < 1/2$ and

$$\frac{\delta_n}{n^{-1/2}} \asymp n^{1/2-\xi} \rightarrow \infty.$$

The lower bound is therefore strictly slower than $\text{root-}n$ at P_0° , whereas `prop : oracle – tie – rootn` supplies a $\text{root-}n$ interval when only the exact-tie stratum must be covered. $\square$

Justification. This is the focal causal least-favorable calculation: a sparse randomized-sign mixture, not the routine Cai–Low two-point transfer, creates the strict adaptation penalty. *Gap.* Cai–Low provide the standard expected-length testing reduction for Gaussian linear functionals; the new target is the explicit causal sparse near-tie mixture, its product chi-square calculation, and its strict cross-stratum penalty. Luedtke–van der Laan, Wu–Wang, Shi–Lu–Song, Xu–Guo, Whitehouse–Chen–Austern–Syrkanis, Ponomarev–Semenova, and Liu–Liu–Sasaki–Wan do not give the displayed two-sided union-honest expected-length lower bound from this explicit dominated causal sparse randomized-sign mixture. *Consumer.* It determines when a single interval for optimized rilpivirine-versus-efavirenz value must be wider at an exact biomarker-response tie than an oracle tie-only analysis.

Lemma 4 (`lem:local-polynomial-policy-regret`). *There is an explicit cellwise local-polynomial estimator, based on m iid training observations and smoothness s_1 , whose induced rule $\hat{d}_m(x) = 1\{\hat{m}_1(x) \geq \hat{m}_0(x)\}$ satisfies*

$$\sup_{P \in \mathcal{P}_1} E_{P^m} \left[V(P) - E_P\{m_{P, \hat{d}_m(X)}(X) \mid \hat{d}_m\} \right] \leq K m^{-s_1(\alpha+1)/(2s_1+d)}.$$

The constant K depends only on the frozen class parameters. For every $P \in \mathcal{P}_0$, the regret inside the expectation is identically zero for every training realization.

Proof. Put $s = s_1$, $q = \lceil s \rceil - 1$, and let $\phi(u) = (u^k : |k| \leq q)$ be the finite vector of tensor monomials on $[0, 1]^d$. Let

$$k_m = \left\lceil m^{1/(2s+d)} \right\rceil, \quad h_m = k_m^{-1},$$

and partition $\mathcal{X}$ into the k_m^d cubes of side h_m . If Q has lower corner c_Q , write $\phi_Q(x) = \phi((x - c_Q)/h_m)$ on Q . For $a \in \{0, 1\}$, define

$$G_{a,Q} = \frac{2}{m} \sum_{i=1}^m 1\{A_i = a, X_i \in Q\} \phi_Q(X_i) \phi_Q(X_i)^\top,$$

$$H_{a,Q} = \frac{2}{m} \sum_{i=1}^m 1\{A_i = a, X_i \in Q\} Y_i \phi_Q(X_i).$$

Let $M_\phi = \int_{[0,1]^d} \phi(u) \phi(u)^\top du$ and $\lambda_\star = \lambda_{\min}(M_\phi)/4 > 0$. If $\lambda_{\min}(G_{a,Q}) \geq \lambda_\star h_m^d$, set $\hat{b}_{a,Q} = G_{a,Q}^{-1} H_{a,Q}$; otherwise set $\hat{b}_{a,Q} = 0$. For $x \in Q$, define $\hat{m}_a(x)$ by truncating $\phi_Q(x)^\top \hat{b}_{a,Q}$ to $[-\bar{B}, \bar{B}]$. This is a completely sample-measurable estimator.

We first establish its pointwise tail bound uniformly over $P \in \mathcal{P}_1$. Randomization gives

$$E_P G_{a,Q} = \int_Q g_P(x) \phi_Q(x) \phi_Q(x)^\top dx.$$

The density lower bound implies

$$\lambda_{\min}(E_P G_{a,Q}) \geq \frac{1}{2} h_m^d \lambda_{\min}(M_\phi) = 2\lambda_\star h_m^d.$$

For every $m_{P,a} \in C^s(\mathcal{X})$, Taylor's formula on a cube gives a polynomial $p_{a,Q}$ of degree at most q satisfying

$$\sup_{x \in Q} |m_{P,a}(x) - p_{a,Q}(x)| \leq K_1 h_m^s.$$

For integer s , this is the degree- $(s-1)$ Taylor bound using the bounded order- s derivatives; for noninteger s , it is the degree- $\lfloor s \rfloor$ Taylor bound using the stated Hölder modulus. The constants are uniform because the Hölder radius is L_1 .

Let $b_{a,Q}^*$ be the population weighted least-squares coefficient. The preceding approximation, the density bounds, and the population Gram lower bound imply

$$\sup_{x \in Q} |m_{P,a}(x) - \phi_Q(x)^\top b_{a,Q}^*| \leq K_2 h_m^s, \quad \|b_{a,Q}^*\| \leq K_2.$$

For independent bounded variables, scalar Bernstein's inequality reads

$$\Pr\{|m^{-1} \sum_{i=1}^m (Z_i - EZ_i)| > t\} \leq 2 \exp\left\{-\frac{mt^2}{2(\sigma^2 + Mt/3)}\right\}$$

when $|Z_i - EZ_i| \leq M$ and $\text{Var}(Z_i) \leq \sigma^2$. Apply this inequality to the finitely many entries of $G_{a,Q}$ and to the coordinates of

$$\frac{2}{m} \sum_{i=1}^m 1\{A_i = a, X_i \in Q\} \phi_Q(X_i) \{Y_i - \phi_Q(X_i)^\top b_{a,Q}^*\}.$$

Each summand is bounded by a class-dependent constant and has variance at most a constant times h_m^d . A finite-dimensional union bound, the population eigenvalue bound, and the inverse perturbation identity therefore give constants $K_3, k_3 > 0$ such that, for every fixed $x \in Q$,

$$P^m\{|\hat{m}_a(x) - m_{P,a}(x)| > K_3(h_m^s + u)\} \leq K_3 \exp(-k_3 m h_m^d u^2) + K_3 \exp(-k_3 m h_m^d)$$

for $0 < u \leq 1$. Truncation cannot increase the error because the true regression lies in $[-\bar{B}, \bar{B}]$.

Set

$$r_m = h_m^s + (m h_m^d)^{-1/2} \asymp m^{-s/(2s+d)}.$$

The last display and a union bound over the two arms imply, uniformly in x , a sub-Gaussian bound for $|\hat{\Delta}(x) - \Delta_P(x)|/r_m$ on its bounded range. If $\hat{d}_m(x)$ chooses the wrong sign and $\Delta_P(x) \neq 0$, then

$$|\hat{\Delta}(x) - \Delta_P(x)| \geq |\Delta_P(x)|.$$

Split the nonzero contrast range into $0 < |\Delta_P| \leq K_4 r_m$ and the dyadic shells

$$2^j K_4 r_m < |\Delta_P| \leq 2^{j+1} K_4 r_m, \quad j \geq 0.$$

The first region contributes at most $K r_m^{\alpha+1}$ by the margin condition. On shell j , the pointwise tail bound and the margin condition give a contribution at most

$$K r_m^{\alpha+1} 2^{j(\alpha+1)} e^{-k 4^j}.$$

The series over j is finite; the exponentially small Gram-failure term is smaller than every fixed power of r_m . Fubini's theorem consequently yields

$$\sup_{P \in \mathcal{P}_1} E_{P^m} E_{P_X} \left[|\Delta_P(X)| 1\{\widehat{d}_m(X) \neq 1\{\Delta_P(X) \geq 0\}\} \mid \widehat{d}_m \right] \leq K r_m^{\alpha+1}.$$

The inner expectation is exactly the policy-value regret, with zero contribution from the tie set. Since $r_m^{\alpha+1} \asymp m^{-s(\alpha+1)/(2s+d)}$, the claimed near-tie bound follows. On $\mathcal{P}_0$, $m_{P,0} = m_{P,1}$ almost surely, so every rule has the same value and the regret is identically zero. $\square$

Lemma 5 (lem:parametric-tie-length-lower-bound). *There is a constant $c_\gamma > 0$ such that every $(C_n)_n \in \mathfrak{C}^{\text{asy}}(\mathcal{U}, \gamma)$ satisfies*

$$\liminf_{n \rightarrow \infty} n^{1/2} E_{(P_0^\circ)^n} [\text{length}(C_n)] \geq c_\gamma.$$

The base law P_0° belongs to both $\mathcal{P}_0$ and $\mathcal{P}_1$, so this lower bound applies to both stratum-wise frontier criteria.

Proof. Choose a fixed $t > 0$, to be made small below, and put $\mu_n = t/\sqrt{n}$. Define a full-data law $F_{t,n}$ exactly as in the base construction except that, conditionally independently for $a \in \{0, 1\}$, the endpoint-supported potential outcome has mean $E[Y(a) \mid X] = \mu_n$. Thus

$$F_{t,n}\{Y(a) = B \mid X\} = \frac{1 + \mu_n/B}{2}, \quad F_{t,n}\{Y(a) = -B \mid X\} = \frac{1 - \mu_n/B}{2}.$$

For all sufficiently large n , these laws are dominated by ν , have uniform design density and propensity one half, satisfy consistency and exchangeability, have constant regressions $m_0 = m_1 = \mu_n$, and obey

$$\text{Var}(Y(a) \mid X) = B^2 - \mu_n^2 \geq v_0.$$

Their constant regression norms are eventually below both L_0 and L_1 , and the margin event $\{0 < |\Delta| \leq u\}$ is empty. Hence their observed images $P_{t,n}$ belong to both $\mathcal{P}_0$ and $\mathcal{P}_1$, while

$$V(P_{t,n}) - V(P_0^\circ) = \mu_n.$$

Under P_0° , the one-observation likelihood ratio is

$$\frac{dP_{t,n}}{dP_0^\circ}(O) = 1 + \frac{\mu_n Y}{B^2}.$$

Therefore

$$1 + \chi^2(P_{t,n}^n, (P_0^\circ)^n) = \left(1 + \frac{\mu_n^2}{B^2}\right)^n \leq \exp(t^2/B^2).$$

Choose $t > 0$ so small that

$$\rho_t := \frac{1}{2} \sqrt{e^{t^2/B^2} - 1} < 1 - 2\gamma.$$

Then the total variation distance between the two product laws is at most ρ_t .

For any union-honest interval sequence, uniform honesty applied to the base and the triangular sequence $P_{t,n}$ gives coverage at least $1 - \gamma - r_n$ for some $r_n \downarrow 0$. Transferring the alternative coverage event to the base law and intersecting it with base coverage yields

$$(P_0^\circ)^n \{0 \in C_n, \mu_n \in C_n\} \geq 1 - 2\gamma - 2r_n - \rho_t.$$

Connectedness forces length at least μ_n on this event. Consequently

$$\liminf_n n^{1/2} E_{(P_0^\circ)^n} [\text{length}(C_n)] \geq t(1 - 2\gamma - \rho_t) =: c_\gamma > 0.$$

The final stratum-wise assertion follows from lem : nonempty – strata, which places P_0° in both observed strata. $\square$

Theorem 2 (thm:split-sample-union-upper-bracket). *Let $\eta = s_1(\alpha + 1)/(2s_1 + d)$. There is an explicit split-sample local-polynomial interval sequence $(C_n^\dagger)_n \in \mathfrak{C}^{\text{asy}}(\mathcal{U}, \gamma)$ such that*

$$\sup_{P \in \mathcal{P}_j} E_{P^n}[\text{length}(C_n^\dagger)] \leq K_\gamma \{n^{-1/2} + n^{-\eta}\}, \quad j \in \{0, 1\}.$$

The same interval is uniformly honest on the near-tie stratum. For every admissible pair, $s_1\alpha < d$ and hence $\eta < \xi$; consequently this proved generic upper bound does not attain the sparse lower scale $\delta_n \asymp n^{-\xi}$.

Proof. Use a deterministic split $I_{\text{tr}} \cup I_{\text{ev}} = \{1, \dots, n\}$, with both sizes asymptotic to $n/2$. On the training sample, form the explicit cellwise local-polynomial regressions $\hat{m}_0, \hat{m}_1$ from lem : local – polynomial – policy – regret, and set

$$\hat{d}(x) = 1\{\hat{m}_1(x) \geq \hat{m}_0(x)\}.$$

For a rule d , write

$$V_P(d) = E_P[m_{P,d(X)}(X)], \quad R_P(d) = V(P) - V_P(d) \geq 0.$$

The lemma gives a constant K_0 , uniform over $P \in \mathcal{P}_1$, such that

$$E_{P|I_{\text{tr}}}|R_P(\hat{d})| \leq K_0|I_{\text{tr}}|^{-\eta}.$$

For $P \in \mathcal{P}_0$, the exact tie gives $R_P(d) = 0$ for every measurable rule d , so the same inequality is valid there.

On the independent evaluation sample define

$$\hat{V}_n = \frac{1}{|I_{\text{ev}}|} \sum_{i \in I_{\text{ev}}} 21\{A_i = \hat{d}(X_i)\}Y_i.$$

Conditional on the training sample, randomization gives

$$E_P[\hat{V}_n \mid I_{\text{tr}}] = V_P(\hat{d}).$$

Each summand lies in $[-2B, 2B]$. Hence, with

$$t_n = B\sqrt{\frac{8\log(4/\gamma)}{|I_{\text{ev}}|}},$$

Hoeffding's inequality gives

$$P^n\{|\hat{V}_n - V_P(\hat{d})| > t_n \mid I_{\text{tr}}\} \leq \frac{\gamma}{2}.$$

Markov's inequality and the regret bound, with

$$b_n = \frac{2K_0}{\gamma}|I_{\text{tr}}|^{-\eta},$$

give $P^n\{R_P(\hat{d}) > b_n\} \leq \gamma/2$, uniformly over $P \in \mathcal{U}$. Define the connected interval

$$C_n^\dagger = [\hat{V}_n - t_n, \hat{V}_n + t_n + b_n].$$

On the intersection of the two preceding events,

$$\hat{V}_n - t_n \leq V_P(\hat{d}) \leq V(P) = V_P(\hat{d}) + R_P(\hat{d}) \leq \hat{V}_n + t_n + b_n.$$

Therefore $\inf_{P \in \mathcal{U}} P^n\{V(P) \in C_n^\dagger\} \geq 1 - \gamma$ for all sufficiently large n , and the interval sequence belongs to $\mathfrak{C}^{\text{asy}}(\mathcal{U}, \gamma)$. Its length is deterministic:

$$\text{length}(C_n^\dagger) = 2t_n + b_n \leq K_\gamma \{n^{-1/2} + n^{-\eta}\}.$$

This proves both displayed stratum-wise bounds.

Finally,

$$\xi - \eta = \frac{s_1(\alpha + 1)(d - s_1\alpha)}{(d + s_1\alpha + 4s_1)(2s_1 + d)}.$$

Admissibility gives $s_1\alpha < 1 \leq d$, so the numerator is positive. Thus $\eta < \xi$, which is the stated honest loose upper bracket rather than a matching $O(\delta_n)$ conclusion. $\square$

Theorem 3 (thm:honest-adaptation-phase-bracket). *Let $\eta = s_1(\alpha + 1)/(2s_1 + d)$. For both $j \in \{0, 1\}$, the honest expected-length frontier has the following proved phase diagram. If $3s_1\alpha < d$, then*

$$0 < \mathfrak{R}_j^{\text{asy}}((n^{-\xi})_n) \quad \text{and} \quad \mathfrak{R}_j^{\text{asy}}((n^{-\eta})_n) < \infty.$$

If $3s_1\alpha \geq d$, then

$$0 < \mathfrak{R}_j^{\text{asy}}((n^{-1/2})_n) \quad \text{and} \quad \mathfrak{R}_j^{\text{asy}}((n^{-\min\{1/2, \eta\}})_n) < \infty.$$

In particular, when $2s_1\alpha \geq d$, a root- n union-honest connected interval is attainable and the root- n order is minimax. In the regimes $3s_1\alpha < d$ and $d/3 \leq s_1\alpha < d/2$, respectively, the proved lower and upper exponents do not coincide, so no matching-frontier conclusion follows from these bounds.

Proof. The upper theorem supplies one interval sequence honest over $\mathcal{U}$ with expected length bounded, on both strata, by

$$K_\gamma\{n^{-1/2} + n^{-\eta}\} = O(n^{-\min\{1/2, \eta\}}).$$

This gives every asserted finiteness conclusion. In the low-margin regime $3s_1\alpha < d$, one has $\xi < 1/2$ and $\eta < \xi$, so the same upper bound is in particular $O(n^{-\eta})$.

For the lower conclusions, lem : nonempty – strata places P_0° in both $\mathcal{P}_0$ and $\mathcal{P}_1$. When $3s_1\alpha < d$, thm : sparse – near – tie – adaptation – lower – bound gives, for every honest interval sequence,

$$\liminf_n \frac{E_{(P_0^\circ)^n}[\text{length}(C_n)]}{n^{-\xi}} > 0.$$

Since the supremum defining either stratum-wise criterion includes P_0° , taking first the stratum supremum and then the infimum over honest sequences proves

$$\mathfrak{R}_j^{\text{asy}}((n^{-\xi})_n) > 0, \quad j \in \{0, 1\}.$$

For every admissible pair, lem : parametric – tie – length – lower – bound similarly gives

$$\mathfrak{R}_j^{\text{asy}}((n^{-1/2})_n) > 0.$$

This is the relevant lower bound when $3s_1\alpha \geq d$, because the displayed sparse separation is then no larger than root- n .

It remains only to identify when the upper exponent is root- n . Direct algebra gives

$$\eta - \frac{1}{2} = \frac{2s_1\alpha - d}{2(2s_1 + d)}.$$

Thus $2s_1\alpha \geq d$ is equivalent to $\eta \geq 1/2$. In that regime the split-sample interval has $O(n^{-1/2})$ length and the parametric lower bound proves minimax optimality. If $3s_1\alpha < d$, the available exponents satisfy $\eta < \xi < 1/2$. If $d/3 \leq s_1\alpha < d/2$, they satisfy $\eta < 1/2$ while the binding proved lower scale is only root- n . These strict inequalities establish the two stated nonmatching brackets without asserting that either gap is sharp. $\square$

Technical internal limitation (diagnostic only)

The proposed multiscale upper route has no established cross-stratum bias bound, so it is not a theorem-level construction. The lower bound uses a deliberately specified two-stratum subproblem: it is binding only when $3s_1\alpha < d$, and it does not itself characterize every tie-mass, margin, and smoothness union.

7 Honest scope

The theorem target is the sparse causal mixture and its strict ordered-pair lower bound under $3s_1\alpha < d$. At $3s_1\alpha = d$, the displayed sparse separation has root- n order; above it, that separation is $o(n^{-1/2})$ and is nonbinding against the tie-only oracle. The two-point expected-length reduction is standard Cai–Low machinery rather than a novelty claim. Attainment by a connected union-honest interval, a stronger obstruction in the nonbinding regimes, extension to arbitrary tie-mass bands, and the global adaptation diagram remain open. The two strata, their radii, the margin parameters, randomization, outcome bounds, and γ are fixed before sampling; unknown continuum smoothness adaptation and monotone coarsening are outside this core.

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